

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech Mid Semester Exam

Semester: 4th

Subject Code: 303191251

Subject Name: Probability, Statistics and Numerical Methods

Date: 31/01/2025

Time: 1hr: 30min

Total Marks: 40

Sr. No.		Marks	Co/Po	Blooms Taxonomy																
Q.1	(A) Five One line Questions 1. Define Positive and Negative correlation. 2. State the equation of Line of Regression of y on x. 3. A card is drawn from a pack of well- shuffled cards. Find the probability of the event that the card drawn is a spade. 4. For which value of k, the given function is a probability mass function? <table><tr><td>x</td><td>0</td><td>1</td><td>2</td></tr><tr><td>$f(x)$</td><td>0.3</td><td>k</td><td>0.2</td></tr></table> 5. Write any one numerical methods to solve transcendental equations?	x	0	1	2	$f(x)$	0.3	k	0.2	05	1 1 2 2 4	Remembering Understanding Applying Analyzing Evaluating								
x	0	1	2																	
$f(x)$	0.3	k	0.2																	
	(B) Five Fill in the blanks 1. When more than two variables are studied excluding some other variables, the relationship is termed as _____. 2. The coefficient of correlation lies between _____ and ____. 3. The expected value of a constant is _____. 4. The probability mass function $f(x)$ Poisson distribution is given by _____. 5. For the Standard normal probability distribution, $\mu =$ _____ and $\sigma =$ _____.	05	1 1 2 2 4	Understanding Analyzing Applying Remembering Creating																
Q.2	Attempt any four. (Short Questions)	12																		
	(1) Draw a scatter diagram indicating the relation between x and y from the following information and discuss about their correlation. <table><tr><td>x</td><td>15</td><td>18</td><td>8</td><td>10</td><td>12</td></tr><tr><td>y</td><td>5</td><td>2</td><td>12</td><td>10</td><td>8</td></tr></table>	x	15	18	8	10	12	y	5	2	12	10	8		1	Creating				
x	15	18	8	10	12															
y	5	2	12	10	8															
	(2) Calculate the coefficient of correlation from the following data: <table><tr><td>x</td><td>12</td><td>9</td><td>8</td><td>10</td><td>11</td><td>13</td><td>7</td></tr><tr><td>y</td><td>14</td><td>8</td><td>6</td><td>9</td><td>11</td><td>12</td><td>3</td></tr></table>	x	12	9	8	10	11	13	7	y	14	8	6	9	11	12	3		1	Applying
x	12	9	8	10	11	13	7													
y	14	8	6	9	11	12	3													

	(3) Suppose that 5 men out of 100 and 25 women out of 10,000 are color blind. A color-blind person is chosen at random. What is the probability of her being female? (Assume males and females to be in equal numbers.)		2	Analyzing																						
	(4) A fair dice is thrown. Find the probability of getting (i) an even number, (ii) a perfect square, and (iii) an integer greater than or equal to 3.		2	Understanding																						
	(5) Solve the following system of linear equations by using Gauss-Seidel method correct up to two decimal places. $8x + y + z = 5$ $x + 8y + z = 5$ $x + y + 8z = 5$		4	Analyzing																						
Q.3	Attempt any two questions	08																								
	(1) Ten participants in a contest are ranked by two judges as follows: <table border="1"><tr><td>x</td><td>1</td><td>3</td><td>7</td><td>5</td><td>4</td><td>6</td><td>2</td><td>10</td><td>9</td><td>8</td></tr><tr><td>y</td><td>3</td><td>1</td><td>4</td><td>5</td><td>6</td><td>9</td><td>7</td><td>8</td><td>10</td><td>2</td></tr></table> Calculate the rank correlation coefficient.	x	1	3	7	5	4	6	2	10	9	8	y	3	1	4	5	6	9	7	8	10	2		1	Applying
x	1	3	7	5	4	6	2	10	9	8																
y	3	1	4	5	6	9	7	8	10	2																
	(2) (a) The probability that a man aged 60 will live up to 70 is 0.65. What is the probability that out of 10 such men now at 60 at least 7 will live up to 70? (b) There are 5 red and 7 black balls in a box. Two balls are drawn at random one after the other. If they are drawn without replacement, find the probability that both the balls are red.		2	Understanding																						
	(3) Find a positive root of the equation $x^3 - 4x + 1 = 0$ by using Regula-Falsi method correct up to three decimal places.		4	Analyzing																						
Q.4	(A) Find the constant k such that the function $f(x) = \begin{cases} kx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$ is a probability density function and compute (i) $P(1 < X < 2)$, (ii) $P(X < 2)$, and (iii) $P(X \geq 2)$.	05	2	Applying																						
	(B) Fit a second-degree polynomial $y = a + bx + cx^2$ using least square method to the following data: <table border="1"><tr><td>x</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>y</td><td>5</td><td>12</td><td>26</td><td>60</td><td>97</td></tr></table> Also, estimate y at $x = 6$.	x	1	2	3	4	5	y	5	12	26	60	97	05	1	Creating										
x	1	2	3	4	5																					
y	5	12	26	60	97																					
	OR																									
	(B) Find the real root of the equation $3x = \cos x + 1$, by using Newton-Raphson method correct up to four decimal places.	05	4	Analyzing																						